



Introduction

Welcome to the MIL electrical team! To help ease you into our lab, we have created this introductory assignment where you will work towards a goal through independent research and design. By the end of this assignment, you will have learned how to use some more essential features of Altium Designer and (potentially) some new PCB design techniques.

Task	Completion (Hours)	Completion (Weeks)
Altium Installation and Tutorial	5	1.5
Introductory Assignment – Research	2 – 3	2 – 3
Introductory Assignment – Schematic	2 – 3	
Introductory Assignment – PCB	2 – 3	
Introductory Assignment – Submission	<1	

Table 1: An estimate of the amount of time needed to complete each assignment task broken down by hours and weeks. The estimate in weeks is based on 3 hours per week for 0 credits of EGN 4912. The initial Altium tutorial is a suggested pre-requisite for this assignment.

Altium Installation and Tutorial

Before delving into the introductory assignment, we recommend going through the initial Altium tutorial that will teach you the basics of navigating through Altium’s menus. You can find the tutorial in the electrical team GitHub organization under the repository *uf-mil-electrical/MIL-Altium/Getting-Started*¹.

Acquiring Altium

It is recommended that you use one of the professional Altium licenses. In the past we have had some issues with acquiring student licenses in a timely manner. To set up an account, reach out to Russell MacGregor, @xicara.de.cafe, on discord and provide the following information:

- A brief request for an Altium license.
- Your name.
- Your UF email address.

While not recommended, you can acquire a free Altium student license from the Altium website². Versions after Altium 19 have had major UI and menu changes which may make it difficult to follow the original tutorial.

1 – Link to the Getting Started repository: <https://github.com/uf-mil-electrical/MIL-Altium/tree/main/Getting%20Started>

2 – Link to the Altium website, student license page: <https://www.altium.com/education/students>



Introductory Assignment – Scenario

As a member of the electrical team, you have done the initial tutorials and assignments and are now looking for a task to get started with. You have chosen the following task from GitHub:

SubjuGator 8 Kill Logic Board

Task Description

Assignees

This task will be worked on by 1 member.

Definitions

This section will define some important terms needed to understand the project:

- **Hardware Kill System:** The hardware-defined external emergency shutoff subsystem. This system operates even when the sub is not tethered. This is connected to the Thruster/Kill board which performs the hard kill action.
- **Software Kill System:** The software-defined (i.e. in ROS) internal emergency shutoff subsystem.
- **Tethering:** The act of connecting the sub's internal network to the shore network by connecting the network cable, or tether, to the sub. After tethering, the software team has access to the sub computer and can perform whatever tasks are needed.
- **Emergency Shutoff:** A term to describe the complete disconnection of the thrusters from the batteries using the thruster/kill board.
- **Hardware Kill Switch:** The interface which divers use to interact with the hardware kill system.

Purpose

The hardware kill system is a safety feature needed to compete in the RoboSub competition and is recommended for practice runs. It allows a diver to immediately stop sub operation in the event of an emergency. The alternative to the hardware kill system is unplugging the batteries which, while safe, is not ideal. This system is typically part of the Thruster/Kill board, which controls the thruster PWM signals and power relays.



Behavior

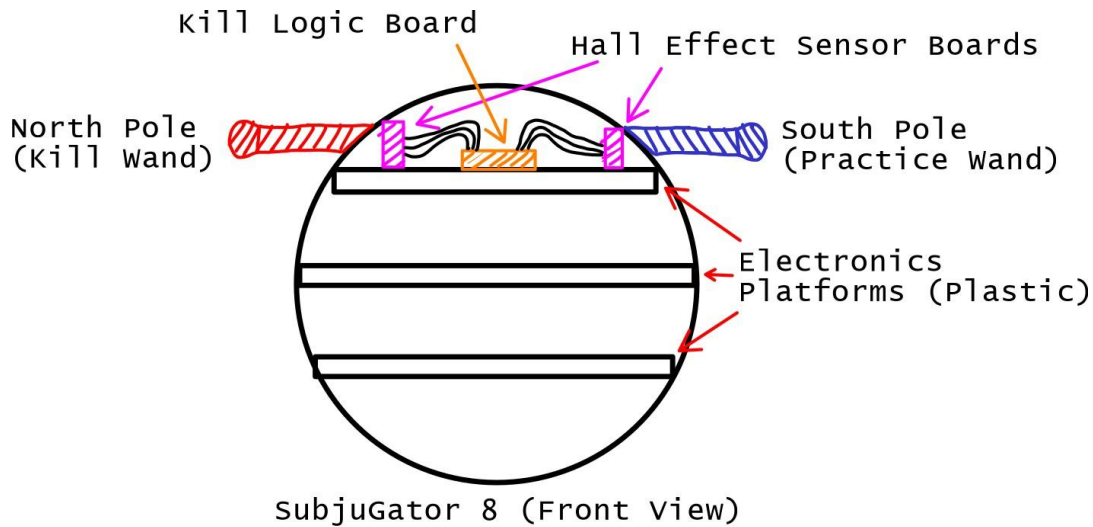


Figure 2: A front view drawing of the hardware kill system on SubjuGator 8.

The current design of the kill system uses two magnet “wands” (PVC pipes with magnets on the ends) that are attached to the sub. One of them is a practice wand which has a south pole magnet, and the other is a kill wand which has a north pole magnet. This system allows testing to be more efficient by making it easy to continue after a hard kill while keeping the kill system safe and simple for competition.

The kill wand’s presence determines whether SubjuGator has been hard killed or not while the practice wand’s presence determines whether the hard kill is reversible or not. If the kill is not reversible SubjuGator needs to be power cycled to continue. The states have been summarized in the following table:

South Pole Magnet	North Pole Magnet	Hard Kill?	Hard Kill Reversible?
Not Present	Not Present	YES	NO
Present	Not Present	YES	YES
Not Present	Present	NO	NO
Present	Present	NO	YES

Table 2: The states for SubjuGator 8’s Hardware Kill System.

The hall effect sensors have already been chosen and made into secondary boards that plug into the main logic board. This allows us to swap out the sensors if we need ones with different sensitivities. Both sensors have open-drain outputs where the signal is grounded while the magnet is present. The state diagram with the sensor information is below:

South Pole (SP)	North Pole (NP)	SP Sensor	NP Sensor	Hard Kill	Reversible Hard Kill
Not Present	Not Present	1	1	1	0
Present	Not Present	0	1	0	1
Not Present	Present	1	0	0	0
Present	Present	0	0	0	0

Table 3: The states for SubjuGator 8’s Hardware Kill System with sensor voltages and kill state represented as Boolean logic. The reversible state is represented as a second, *independent*, output that is sent to the Thruster/Kill board.



Pre-requisites

- Basic Altium Knowledge (Altium tutorial)
- Digital Logic (EEL3701 or equivalent experience with mixed logic)

Task Requirements

Your task is to make the main logic board that follows the logic set in this GitHub issue. The outputs for Hard Kill and Reversible hard kill may have any logic level you want (since they will be connected to a microcontroller).

Size & Layout

- The board should be no more than 2" x 2".
- You may use 2 or 4 layers.
- Ensure that there are 4 M3-sized holes on the corners for mounting the board to SubjuGator.

Make sure to follow standard logic gate guidelines (i.e. tie unused inputs to VCC or GND).

Remember to perform schematic validation and design rule checks. Errors other than silk-to-silk, minimum solder mask sliver, and hole size constraints (for the M3 screw holes) should not exist.

Using your Digital Logic knowledge, you have made the following logic circuit diagram and selected the TC74AC00PF quad 2-input NAND gate³ as the chip you will use:

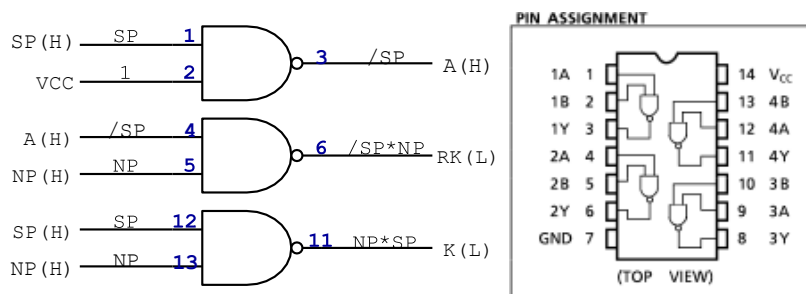


Figure 3: Your labeled logic circuit diagram (left) and a top view of the TC74AC00PF IC from its datasheet (right).

Introductory Assignment – Task

Now that you have decided on a solution, you must make the schematic and PCB for the logic board. You have completed a lot of the work but are missing the NAND chip and its connections and they are not provided in any of the MIL component libraries. You also think it might be a good idea to add LED status lights for the different logic states of the kill system.

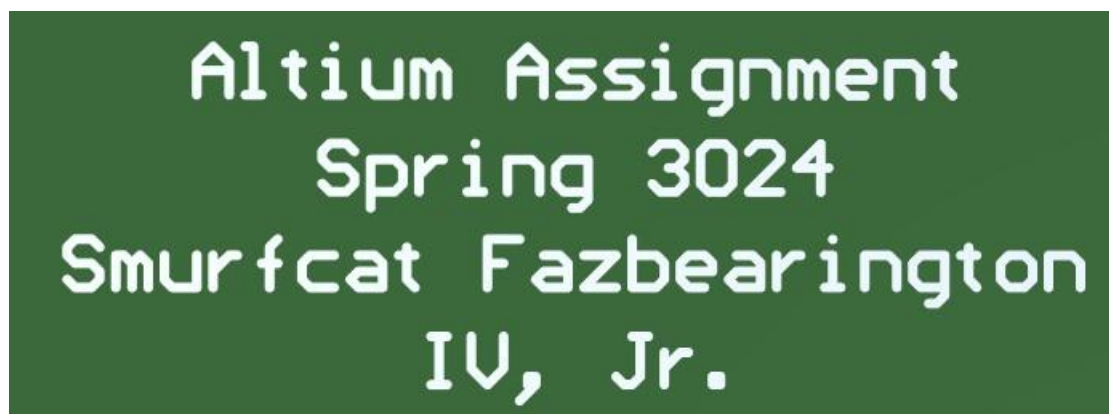
³ <https://www.digikey.com/en/products/detail/toshiba-semiconductor-and-storage/TC74AC00PF/870816>



To make this assignment easier to complete, we have provided an incomplete schematic, PCB, and associated libraries to get you started. [You may find this information under the “Altium Assignment” folder in the “Getting Started” repository⁴.](#)

Deliverables

- Complete Schematic and PCB with:
 - o TC74AC00PF NAND gate:
 - Schematic symbol
 - PCB footprint with associated 3D model
 - o Appropriate bypass capacitors
 - o Valid and complete layout (including keep out layer). I recommend organizing the components on the side so that they have minimal crossovers before you start routing.
 - o Your name in the schematic’s first author bullet point and an attribution on the PCB like figure 3.
 - o Labels for the connectors and at least one of each of their pins (e.g. “South Pole” and “3V3”).
 - o *(Optional)* Rounded corners
 - o *(Optional)* LED status lights with labels:
 - Not killed (green)
 - Reversible hard kill (yellow)
 - Hard kill (red)
 - o *(Optional)* M3 mounting holes (please use the provided Pad/Via library) with size label (“M3”).
- Complete Schematic and PCB library with:
 - o TC74AC00PF NAND gate:
 - Schematic symbol
 - PCB footprint with associated 3D model
- PDF report containing:
 - o Schematic Printout
 - o PCB prints (top layer and bottom layer)
 - o PCB 3D prints (top layer and bottom layer)
 - o Bill of Materials
 - o Design Rules Check
 - o Electrical Rules Check
- Folder containing NC Drill and Gerber production outputs



⁴ <https://github.com/uf-mil-electrical/Getting-Started/tree/master/Altium%20Assignment>



Figure 4: An example attribution for the PCB.

Research Material

The provided research material is not exhaustive; you may perform further research for topics not explained in the list below.

Adding Components and Footprints to Libraries

- <https://youtu.be/PqFtSpAXB9Q>

The timestamps below are for information specific to footprint and schematic component creation/importing. The video has a lot more useful information though!

- o 3:32 – 27:24
- o 30:10 – 49:57

Footprint Wizard

- https://www.altium.com/documentation/altium-designer/creating-pcb-footprint#!creating_footprint_using_footprint_wizard
- <https://www.pcb-3d.com/wordpress/tutorials/how-to-calculate-pt-hole-and-pad-diameter-sizes-according-to-ipc-7251-ipc-2222-and-ipc-2221-standards/>

Unused 74HC Inputs

- <https://electronics.stackexchange.com/questions/384672/74hc-hct-what-to-do-with-unused-inputs-and-why>

Polygon Pour

Note that this is likely not needed if you are doing a 4-layer board design (as polygon pour on the top and bottom layers offer little benefit in this design) but I have provided resources in case they are needed.

- <https://www.altium.com/documentation/altium-designer/pcb-signal-layer-polygons>
- <https://electronics.stackexchange.com/questions/610922/should-i-do-a-ground-copper-pour-on-a-4-layer-pcb-in-order-to-shorten-return-cur>
- <https://youtu.be/R3w4Go1s1hM>

Stitching Vias (Multilayer Boards)

This is likely not needed if you are doing a 4-layer board design (as polygon pour on the top and bottom layers offer little benefit in this design) but I have provided resources in case they are needed.

- <https://resources.altium.com/p/everything-you-need-know-about-stitching-vias>

Editing Schematic Parameters

- <https://www.altium.com/documentation/altium-designer/setting-up-schematic-document#!document-level-parameters>

Using Pad/Via Libraries

- <https://www.altium.com/documentation/altium-designer/pad-via-templates>
- <https://www.youtube.com/watch?v=JEiMSW9xEeg>
(Altium 15, but still relevant as of Altium 23)



Via in Pad Technology

- <https://youtu.be/FM3pRM0CxGw>
- <https://youtu.be/REIR37KW7jg>

Bypass/Decoupling Capacitors

- <https://youtu.be/BcJ6UdDx1vg>
 - o 0:00 – 11:13
- <https://e2e.ti.com/support/logic-group/logic/f/logic-forum/978830/faq-how-do-i-select-a-bypass-capacitor-for-a-cmos-logic-device>
- <https://www.engineering.com/story/decoupling-capacitors-are-crucial-in-pcb-design>
- <https://electronics.stackexchange.com/questions/12286/bypass-capacitors-needed-in-low-frequency-digital-logic-systems>

Altium Outjob Files

We have provided an output job file for you to use for submission but if you are interested in learning how to make your own I have provided the following resources. You could also use the provided file as an example.

- <https://youtu.be/IGJ3OSAXd94>
- <https://youtu.be/kb5kHHwPT40>

Assignment Files

- Incomplete Schematic and PCB
- Incomplete Schematic, and PCB libraries
- Via library (M3 screw holes)
- OutJob file for submission

Tips

- Focus on one piece at a time. First, create the schematic footprint with labeled and named pins, place it on the schematic, wire it up, and check the validity of the schematic. If it is good, then move on to making the PCB footprint and tying it to the existing schematic component.
- If you have placed the NAND gate, make sure to update the schematic component after every change. No changes will be reflected in already placed components otherwise.
- Look at the IC's datasheet for help to create the footprint and find the 3D model. IC packages like these are typically standardized, so you may use the package to find a generic 3D model that represents the NAND gate.
- Sites like GrabCAD could be useful for finding the required 3D models. If you have experience with software like SolidWorks, Autodesk Fusion/Inventor, or OnShape you can create the model yourself from the IC's datasheet or other online sources.
- Some advanced Altium tools (like the footprint wizard) could be useful.

Introductory Assignment – Submission

Once you have completed the assignment, you may use the output job file we have provided with the project to package your submission. Once you press “generate content” for the report and PDF sections, you should have a



new folder named “Assignment Submission” in your project’s directory with the generated files, and a PDF in the project's directory.

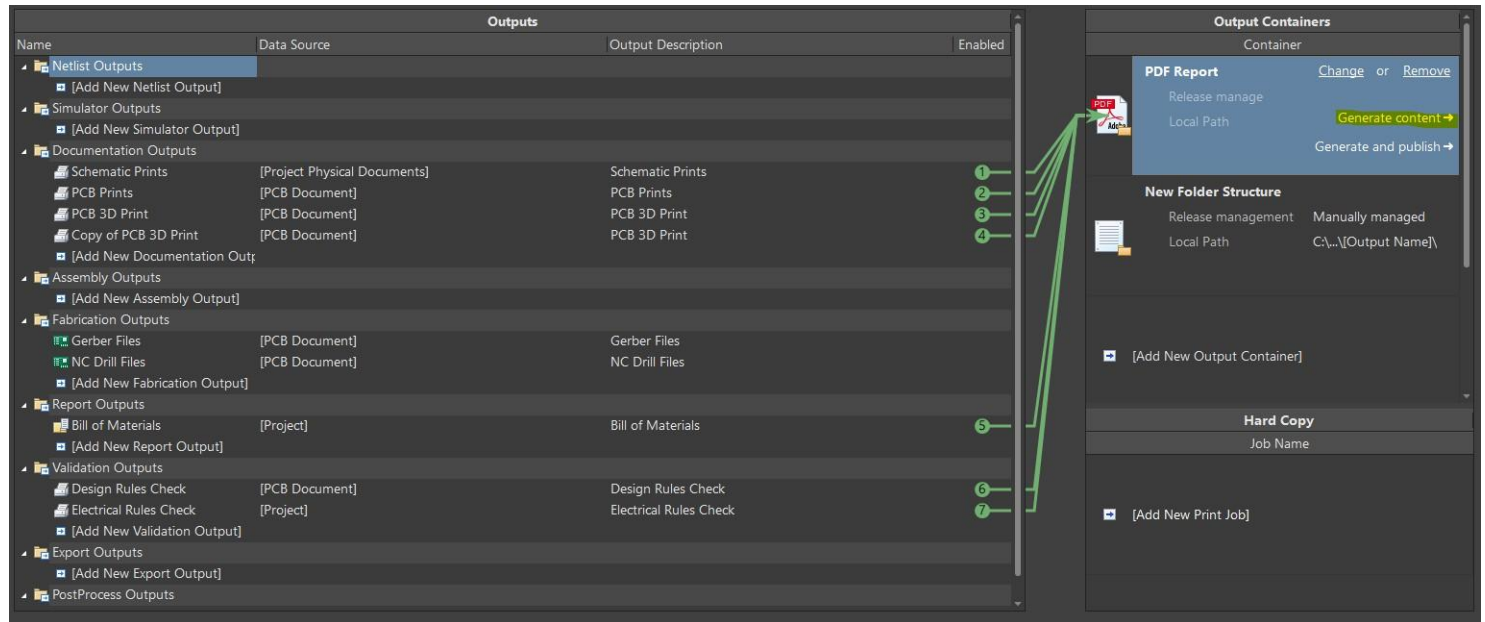


Figure 5: What the output job file looks like with the highlighted “generate content” button.

Please add your name to the PDF file name so that it reads:

[Last Name][First Name]_IntroductoryAssignmentReport.PDF
(e.g. **SmithJohn_IntroductoryAssignmentReport.PDF**)

Please create a **.zip** file from everything in the project directory excluding the PDF file and title it:

[Last Name][First Name]_IntroductoryAssignmentFiles.zip
(e.g. **SmithJohn_IntroductoryAssignmentFiles.zip**)

You will submit this compressed file along with the PDF report.